

CALCULATION OF AIRCRAFT NOISE DURATION

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This paper considers the problem of calculating the effective noise duration at any specific observation point during an aircraft fly-past. Existing proposals are reviewed and the results of alternative methods of calculation are compared with each other and with analyses of actual noise history recordings.

1. INTRODUCTION

It is generally acknowledged at present that, as an aircraft flies past, the maximum weighted noise level† experienced or measured by an observer does not give sufficient indication of the “noisiness” of the event. Allowances also have to be made for noise duration and other factors such as pure tones in the noise spectrum.

The method whereby an allowance for noise duration should be made has been the subject of considerable controversy, but the concept that the allowance should be based on the time-integrated value of the instantaneous (weighted) noise intensity appears to have gained general acceptance [1, 2]. The elegance of this concept is attractive but, unfortunately, it gives rise to a number of problems which have not yet been solved satisfactorily. For instance, it is not practically feasible to integrate the recorded noise histories of a large number of fly-pasts when the noise around an airport is being assessed. The limits of integration are also very difficult to determine with any accuracy when analysing a recorded noise history. The result is that the existing formulae for computing a noise duration allowance in this way rely on empirical factors of dubious or limited validity. These empirical factors are often disguised in a cloak of respectability as is, for instance, the case with the “10 dB-down” procedure proposed by the International Organization for Standardization (ISO) [2] and incorporated in a recent report of the International Civil Aviation Organization (ICAO) [1].

A possible solution to the problem of integrating an impracticably large number of noise-time histories is to compute an effective noise duration where, ideally, the product of this duration and the maximum instantaneous noise intensity would be equal to the time-integrated value of the instantaneous noise intensity. If this could be done in terms of simple variables such as aircraft speed and the distance between the aircraft and the observer, the necessity for doing a large number of integrations would be avoided. Unfortunately, all attempts in this direction have, so far, resulted in the same dubious empiricism mentioned above.

The South African method [3] of assessing aircraft noise around airports has, until recently, also made use of noise-duration formulae which incorporate a great deal of empiricism. However, further examination of the principles involved has since pointed the way to more satisfactory methods as will be described in subsequent sections of this paper.

2. REVIEW OF EXISTING METHODS

A duration allowance, in the case where an effective noise duration is explicitly calculated, generally takes the form of an additive allowance defined by

$$\Delta = A \log(T/T_0), \quad (1)$$

† For the purpose of this report, it is assumed that the noise level is measured or calculated in terms of PNdB, dB(A), dB(D) or any other suitable units.

where A is a constant, T is an effective noise duration, and T_0 is an arbitrary constant having the dimensions of time.

In order to conform with the generally accepted energy concept discussed in section 1, the value $A = 10$ should be used, although other values have also been proposed in the past [4].

A considerable variety of methods have, however, been used to calculate the effective noise duration T .

An early American study [5] of military aircraft noise yielded results which suggested that duration would generally be given with sufficient accuracy by the formula

$$T = s'/220 \text{ s}, \quad (2)$$

where s' = shortest distance between observer and flight path, in feet, and where the formula applies to aircraft flying at speeds within the range 150–350 knots (78–180 m s⁻¹). If the constant 220 is considered to represent an average speed, which means that it should be given in knots, then the formula might be rewritten as follows:

$$T = 1.3s'/u \text{ s}, \quad (3)$$

where s' = shortest distance between observer and flight path, in m, and u = aircraft speed, in m s⁻¹.

A German study [6] of larger jet transport noise, as well as later American work of a similar nature, yielded results which led to the relationship

$$T = 3.4s'/u \text{ s}. \quad (4)$$

Early South African work [7] on combat jet aircraft yielded

$$T = 0.78s'/u \text{ s}. \quad (5)$$

Later South African investigations [3] found that the difference between equations (3), (4) and (5) could be accounted for by introducing aircraft mass as a variable. A term to account for ground attenuation was also introduced, thus yielding

$$T = 10(W/3 \times 10^4)^{0.5} e^{-u/170} (s'/300)^{0.8(1-e^{-\gamma})}, \quad (6)$$

where W = aircraft mass in kg, and γ = angle between the line of maximum noise radiation (joining the aircraft and the observer) and the ground, in degrees.

The most recent proposal for calculating an allowance for noise duration is contained in documents of the ICAO [1] and is based on a draft ISO recommendation [2] which specifies the addition of a duration allowance defined by

$$\Delta = 10 \log(1/T_0) \int_{-\infty}^{\infty} 10^{L_N/10} dt - L_{N\max}, \quad (7)^\dagger$$

where L_N = tone-corrected instantaneous perceived noise level and T_0 = an arbitrary constant having the dimensions of time.

In view of what has been said earlier, it is obvious that equations (2) to (5) are admirable examples of the art of empiricism, or as Robinson [4] puts it: "There is as much mythology as science in this area." It is, in fact, easy to understand why there are such significant differences between these equations because in some cases the "10 dB-down" method‡ was used to determine the "actual" noise duration, whereas in other cases attempts were made to

† Equation (7) may also be expressed as

$$\begin{aligned} \Delta &= 10 \log(1/T_0) T 10^{L_{N\max}/10} - L_{N\max} \\ &= 10 \log(T/T_0). \end{aligned}$$

‡ The noise duration is taken to be equal to the time interval over which the instantaneous weighted noise level is within 10 dB of its peak value.

utilize the time-integrated value of the instantaneous noise intensity. In addition, the conditions regarding type of aircraft, ambient noise level, etc., differed considerably.

Of more concern to this discussion is the fact that an equation of the form

$$T = Ks'/u,$$

where K = a constant, must clearly be of limited applicability. Such an equation predicts ever-increasing noise durations as the distance increases, even though the aircraft may be so far removed from the observer that it cannot be heard above the ambient noise level. Conversely, this form of equation predicts long noise durations even when the ambient noise level is extremely high. It therefore disregards the effect of ambient noise level completely, and some arbitrary limit on maximum values usually has to be imposed.

Equation (6) is a more sophisticated version of the same type of equation. By introducing aircraft weight as a parameter, it makes allowance for the fact that the higher noise levels of large aircraft are likely to be heard for longer periods than the lower noise levels of lighter aircraft. It also introduces a factor $(1 - e^{-\gamma})$ to account for ground attenuation and generally provides answers which correlate well with the values predicted by equations (2)–(5). Nevertheless, it does not manage to overcome the objection mentioned in the previous paragraph and must therefore be regarded as unsatisfactory.

Equation (7) avoids the explicit calculation of an effective noise duration but requires integration of each individually measured noise history. This may be a feasible procedure to follow during noise certification tests on a single aircraft but not when the noise climate around airports is being assessed. The ICAO version of this formula further introduces a “working” concept of the limits of integration, namely, that $(t_2 - t_1)$ be taken as the lesser of the two time intervals during which the instantaneous value of the (tone-corrected) perceived noise level is either greater than 90 PNdB or within 10 dB of the maximum level. The introduction of the 90 PNdB condition does mean that the duration allowance becomes zero when the theoretical maximum perceived noise level of the aircraft falls below 90 PNdB but neither the value 90 nor the value 10 can claim much theoretical justification. They are, in effect, equivalent to empirical constants of limited validity.

It is clear, therefore, that existing procedures for calculating the time-integrated value of the instantaneous noise intensity, or the effective noise duration, leave some room for improvement. Alternative approaches to the problem will, therefore, be considered in the next section.

3. THEORETICAL ANALYSIS

3.1. DEFINITIONS

As discussed in section 1, an effective noise duration T may be calculated such that

$$T10^{L/10} = \int_{-\infty}^{+\infty} 10^{L_i/10} dt,$$

or

$$T = 10^{-L/10} \int_{-\infty}^{+\infty} 10^{L_i/10} dt \quad (8)$$

where L_i = instantaneous noise level, L = maximum value of L_i , and t = time. In order to determine finite limits of integration for equation (8), it is necessary to consider a graphical interpretation of the situation.

Figure 1 shows the qualitative relationship, at some point of observation, between the noise level attributable to a specific source (hereafter referred to as the source noise level),

the ambient noise level and the overall noise level. It is clear that L_i may be taken as being either the overall noise level or the source noise level.† If the former is assumed, the limits of integration can conveniently be determined by arbitrarily defining these limits as being the intersections of the overall noise level graph with a horizontal line which represents a noise level of, say, 3 dB higher than the ambient noise level (points A and B in Figure 1).‡

A practically identical and equally rational result is obtained if it is assumed that L_i is equal to the source noise level and that the limits of integration are defined by the intersections of the source noise level graph with the horizontal line representing the ambient noise level

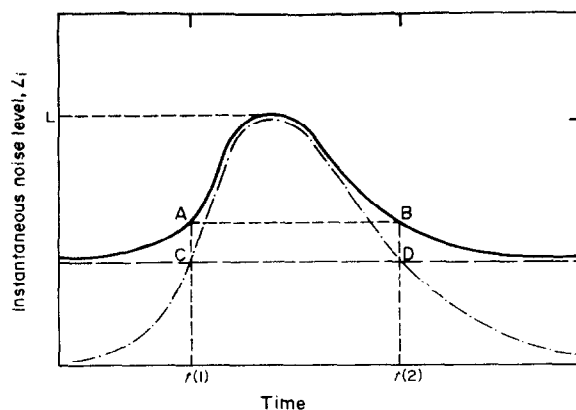


Figure 1. Relationship between source noise level (---), ambient noise level (—) and overall noise level (—).

noise level (points C and D in Figure 1). This assumption has the additional advantage that it simplifies the mathematical treatment of the problem. Consequently, for the purpose of this report, an effective noise duration is defined as

$$T = 10^{-L/10} \int_{t(1)}^{t(2)} 10^{L_i/10} dt \quad (9)$$

where L_i = instantaneous noise level attributable to a specific source, L = maximum value of L_i , and $t(2) - t(1)$ = period during which L_i exceeds the ambient noise level.

3.2. COMPUTER MODEL

The South African method of aircraft noise assessment provides an equation whereby the noise level in dB(A) or dB(D) at any distance from an aircraft may be calculated theoretically.§ This equation is of the form

$$L_i = L_p - \Delta L - \Delta L_s - \Delta L_a - \Delta L_g, \quad (10)$$

where L_i = noise level at the point of observation, L_p = reference polar noise level at a distance of 100 m from the aircraft, ΔL = difference in noise level due to power and speed variations,

† The ICAO formula [equation (7)] avoids making this distinction by confining itself to that portion of the noise-time history where the source noise level is approximately equal to the overall noise level. Other empirical formulae [equations (2) to (5)] appear to have considered the overall noise level only.

‡ There appears to be some philosophical justification for this assumption because it may be argued that a noise will probably not arrest attention unless it causes the overall (ambient) level to increase by a certain minimum number of dB. What this increment should be is, of course, open to argument, but the assumption of a 3 dB increment does not appear to be unreasonable.

§ The level in PNdB may also be calculated but the computation procedure is considerably more involved than the relatively simple method described here.

ΔL_s = difference in noise level due to distance, ΔL_a = difference in noise level due to air attenuation, and ΔL_g = difference in noise level due to ground attenuation.

Provided, therefore, that the reference noise level is available in the form of a polar diagram of the noise radiation pattern around an aircraft, the level L_i (at any instant and at a specific observation point) may be calculated for any particular flight path. The mathematical details of this computation are given in Appendix I.

The effective noise duration for a particular set of conditions can therefore be obtained by calculating L_i with respect to time, and subsequently evaluating equation (9) by means of a computer. An example of such a computation is presented in Figure 2 which shows the

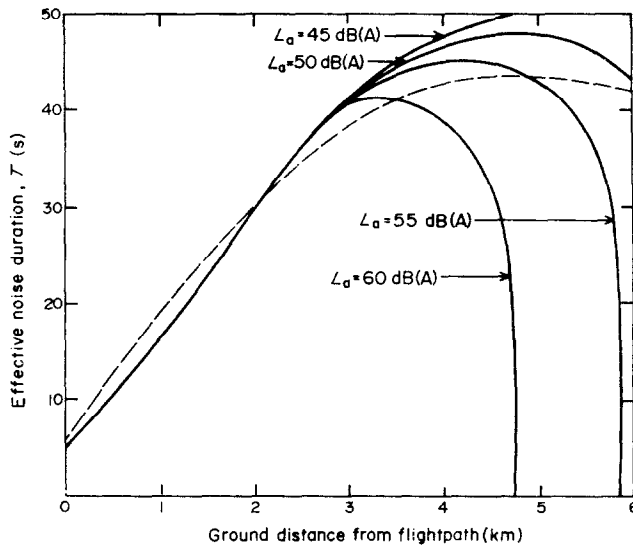


Figure 2. Computer solutions of effective noise duration for different values of the ambient noise level, L_a (—), empirical solution (6) (---). Medium range jet transport at a speed of 120 m s^{-1} and an altitude of 300 m.

effective noise durations at various ground distances from the flight path. The results apply to a medium-range jet transport aircraft flying at an altitude of 300 m with a speed of 120 m s^{-1} . By way of comparison, the results obtained from the empirical equation (6) for similar conditions are plotted as a dotted line in the figure.

The empirical and computer solutions agree closely at moderate ground distances from the flight path. At fairly large ground distances, on the other hand, the empirical equation disregards the effect of ambient noise level completely, whereas the computed results are highly sensitive to this factor. It is obvious that the omission of ambient noise level in the calculation of effective noise duration may, therefore, result in large errors.

3.3. APPROXIMATE MODEL

A detailed computation of L_i in terms of time to obtain the effective noise duration may be avoided by assuming that the noise history graph is symmetrical about $t = 0$, where $L_i = L_a$.† This assumption enables equation (9) to be re-written as

$$T = 2 \left\{ 10^{-L/10} \int_0^{t_a} 10^{L_t/10} dt \right\}, \quad (11)$$

where t_a = time value at which $L_i = L_a$, and L_a = ambient noise level.

† Appendix II discusses the validity of this assumption in more detail.

Equation (11) may be integrated directly, provided that a suitable equation for the instantaneous noise level L_i in terms of time can be found. Since there is a good deal of similarity between a noise-time history and a probability distribution, it is relatively easy to deduce that, for $t > 0$, a suitable relationship may be of the form

$$10^{L_i/10} = 10^{L/10} e^{-a(ut/s')^b}, \quad (12)$$

where L = maximum value of L_i , a, b = empirical constants, u = aircraft speed, t = time, and s' = shortest distance to flight path.

The values of a and b were determined by means of a logarithmic plot of equation (12) as shown in Figure 3, the value of b being given by the slope of the regression lines and the value of $\log a$ by the intercepts on the vertical axis.† The points on the graph were obtained from

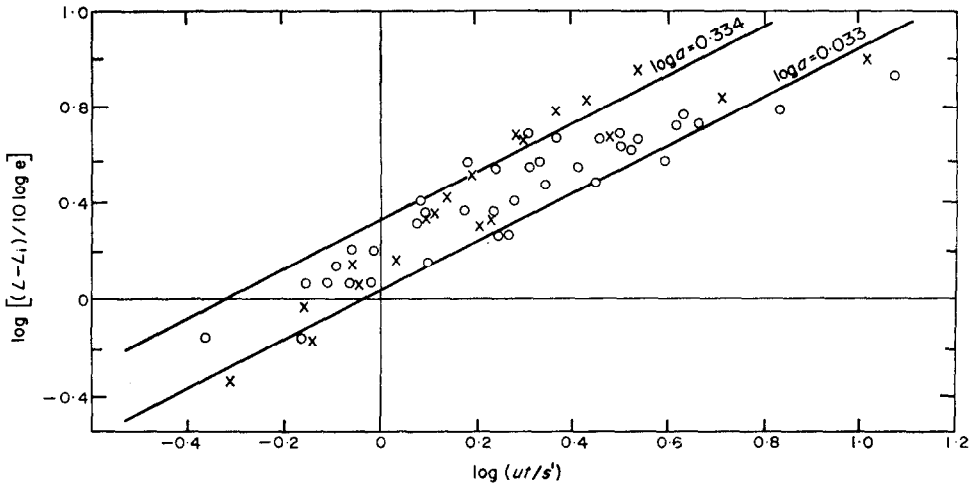


Figure 3. Determination of empirical constants a, b in the equation for instantaneous noise level. $\circ$, Measured values; $\times$, computer results.

actual noise histories recorded during take-off and fly-past operations of a variety of turbojet transport aircraft and from the results of the computer simulation described in section 3.2.

From the Figure it appears that b may conveniently be taken to be equal to 1 but that $\log a$ varies between 0.033 and 0.334. This variation in the value of a is apparently associated with the amount of ground attenuation involved because the higher values of $\log[(L - L_i)/10 \log_{10} e]$ mostly occur at small values of the angle γ between the line of maximum noise radiation and the ground surface.

Substitution of equation (12) in equation (11) yields the following results:

$$T = \frac{2s'}{u} \int_0^{t_a} e^{-a(ut/s')} d\left(\frac{ut}{s'}\right), \quad (13)$$

where, from equation (12),

$$t_a = \frac{s'}{u} \left(\frac{L - L_a}{10a \log_{10} e} \right). \quad (14)$$

Integration of equation (13) then yields

$$T = \frac{2s'}{ua} \left\{ 1 - e^{-(L - L_a)/4.34} \right\}. \quad (15)$$

† The mathematical details of the determination of a and b are discussed in Appendix IV.

The variation in the value of a with changes in the value of γ can be accounted for by writing†

$$a = 1.08(1 + e^{-\gamma/2}), \quad (16)$$

so that, finally,

$$T = \frac{1.85s'}{u(1 + e^{-\gamma/2})} \{1 - e^{-(L-L_a)/4.34}\}. \quad (17)$$

A qualitative indication of the validity of equation (17) may be obtained in a number of ways. For example, for large values of γ and $(L - L_a)$, equation (17) reduces to

$$T = 1.85s'/u. \quad (18)$$

It is clear that this equation provides results which fall within the same range as those of the previously proposed equations (2)–(5).

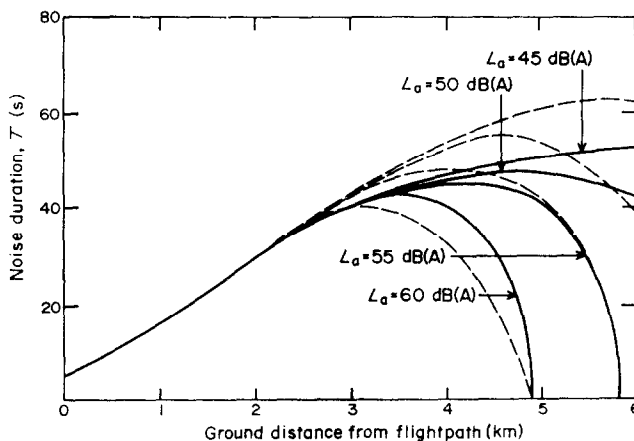


Figure 4. Noise durations at various ground distances from the flight path of a medium jet transport aircraft. —, Computer results; ---, approximate results [equation (17)]. Aircraft speed = 120 m s^{-1} ; aircraft altitude = 300 m .

A more rigorous test of equation (17) is, however, made in Figure 4 which shows that close agreement is obtained with the results of the computer simulation. The validity of the computer simulation is, in turn, substantiated by the close agreement obtained between the computer results and the results of the actual recorded noise histories (Figure 3), as well as the results obtained by means of equation (6) (Figure 2).‡

It may therefore be deduced that the calculation of effective noise duration by means of equation (17) is not only relatively simple but also reliable. In addition, it largely overcomes the inaccuracies which are likely to intrude when the ambient noise level is neglected in such calculations.

4. CONCLUSIONS

A formula [equation (17)] has been developed which enables the effective noise duration during an aircraft fly-over to be calculated from a knowledge of the following variables: (i) the distance between the observer and the flight path; (ii) the aircraft speed at the instant of maximum noise generation; (iii) the maximum noise level experienced by the observer;

† For large γ , $a = 1.08$ and $\log a = 0.033$. For $\gamma = 0$, $a = 2.16$ and $\log a = 0.334$. The factor 2 in the parameter $e^{-\gamma/2}$ was found to provide good correlation with actual measured values.

‡ It should be noted that equation (6) was developed on the basis of all the available theoretical and experimental results existing at the time and that it provides a reasonable degree of correlation between equations (2)–(5).

(iv) the angle between the directional line of maximum noise radiation and the horizontal at the instant of maximum noise generation; and (v) the ambient noise level.

At moderate distances from the flight path this formula provides approximately the same answer as would be obtained from the previous formula employed in the South African noise assessment procedure. There are, however, significant differences compared with the well-known 10 dB-down procedure.† At larger distances from the flight path the formula provides results which are considerably more realistic than those yielded by previous procedures. Some comparative figures are presented in Appendix V. The use of controversial parameters such as aircraft mass in the calculation, and the errors resulting therefrom, are also avoided.

The variables listed above are normally available from other calculations performed in the noise assessment procedure, with the exception of the ambient noise level. The latter depends on actual environmental conditions, but, because these vary considerably with time, it may be quite difficult to decide on suitable values. Table A1 in Appendix III may provide some guidance in this respect.

It should also be noted that the formula for calculating effective noise duration, as developed in this paper, is subject to certain limitations. First, it assumes that the aircraft speed remains fairly constant. At moderate distances from the flight path the assumption is sufficiently accurate since only a relatively short portion of the flight path is involved during the generation of a noise history. At long distances from the flight path it will be less accurate but not to such an extent as to cause very significant errors unless the aircraft speed changes very rapidly. Second, the formula does not give realistic duration values during the early part of the ground run when the aircraft is stationary or travels very slowly. This limitation may be overcome by using a speed of, say, 50% of the lift-off speed whenever the actual speed is less than this; this will provide fairly realistic answers. Third, the formula does not provide reliable answers at certain observation points, e.g. in the vicinity of the centre of curvature when the aircraft executes a sharp turn. The best solution in such a case is to discard the concept of effective noise duration completely and to revert to a detailed calculation of the time-integrated noise energy. Fourth, the formula does not apply to VTOL, STOL or supersonic aircraft.

Despite the above limitations, it is felt that the formula proposed in this paper provides a number of advantages in the computation of the noisiness experienced during aircraft fly-pasts and that it is an improvement on methods hitherto employed.

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† During the development of the previous South African formula [equation (6)], duration results obtained by means of the 10 dB-down procedure were, amongst others, employed in the derivation of the formula.

APPENDIX I

EQUATIONS [3] FOR CALCULATING NOISE-TIME HISTORY BY MEANS OF A COMPUTER

If attenuation due to power and speed variations is neglected, the noise level in dB(A) or dB(D) at any distance from an aircraft is given by

$$L_i = L_p - 20 \log \frac{s_i}{s_0} - \epsilon \left(\frac{s_i - s_0}{s_0} \right) - \psi \log \frac{s_i}{s_0} - \eta \left(7 + 10 \log \frac{s_i}{s_0} \right) e^{-\gamma/2}, \quad (\text{A1})$$

$$\sin \gamma = H/s_i, \quad (\text{A2})$$

$$s_i = + (d^2 + H^2 + u^2 t^2)^{1/2}, \quad (\text{A3})$$

where s_i = distance between the observer and the aircraft (m), s_0 = reference distance (100 m), L_p = reference polar noise level [dB(A) or dB(D)], ϵ = attenuation per 100 m in the one-third octave band with centre frequency of 500 Hz (dB) [1], $\psi = 0$ for take-off, 4 for landing, $\eta = 1.0$ for level terrain with grass up to 300 mm height (for other values, see reference [3]), H = aircraft altitude (m), d = shortest ground distance between the flight path and the observer (m), u = aircraft speed (m s^{-1}), and t = time (s); taken to be zero when the aircraft is opposite the observer.†

Since the reference polar noise level L_p assumes different values at different points around the aircraft, these values together with the corresponding values of the polar angle ϕ_i must be read into the computer. The value of the angle ϕ_i at any instant is calculated from

$$\cos \phi_i = (d^2 + H^2)^{1/2}/s_i. \quad (\text{A4})$$

APPENDIX II

THE SYMMETRY OF NOISE-HISTORY GRAPHS

Figure A1 presents a plot of the polar noise level L_p in dB(A) at a distance of 100 m from a medium-range jet transport aircraft in terms of the angle ϕ_i , where ϕ_i is the angle between a line perpendicular to the aircraft axis and the line joining the aircraft and the observer. ($\phi_i = 90^\circ$ represents a point behind the aircraft and in line with the aircraft axis.)

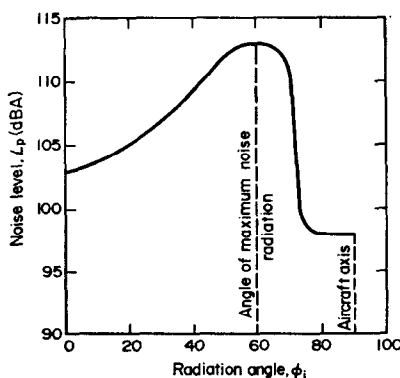


Figure A1. Noise radiation pattern.

† In the case of the approximate model (section 3.3), $t = 0$ when $L_i = L$.

A superficial inspection would seem to indicate that, as an aircraft flies past an observer, the noise level at the observer should increase gradually until $\phi_1 = 60^\circ$ after which it should decrease abruptly. However, this is incorrect because the influence of the noise radiation pattern on the noise-time history at the observer is represented by the mathematical expression

$$\frac{\delta L_p}{\delta \phi_1} \cdot \frac{d\phi_1}{dt}.$$

Although $\delta L_p / \delta \phi_1$ is relatively small for $\phi_1 < 60^\circ$ and relatively large for $\phi_1 > 60^\circ$, $d\phi_1/dt$ follows the opposite trend with the result that

$$\left| \frac{\delta L_p}{\delta \phi_1} \cdot \frac{d\phi_1}{dt} \right|_{\phi_1 < 60^\circ} \simeq \left| \frac{\delta L_p}{\delta \phi_1} \cdot \frac{d\phi_1}{dt} \right|_{\phi_1 > 60^\circ}.$$

Consequently, the rate of increase of the noise level at the observer is roughly equal to the rate of decrease of the noise level, i.e. it may be assumed that the noise-time history is approximately symmetrical about $t = 0$. In cases where the maximum noise radiation angle is small or equal to zero it is usually found that the noise radiation pattern itself is fairly symmetrical.

Figure A2 presents a noise-time history, computed according to the method discussed in section 3.2 which provides further substantiation of the assumption.

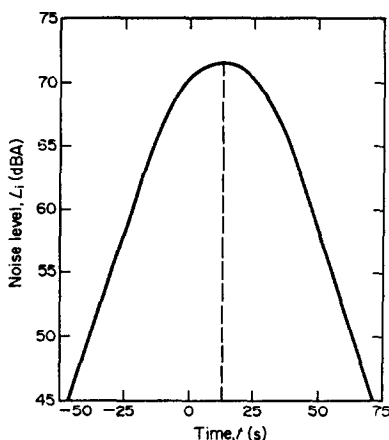


Figure A2. Computer solution of noise-level time history. Medium range jet transport at an altitude of 300 m, ground distance to flight path = 3 km, aircraft speed = 120 m s^{-1} .

It should, however, be noted that the above argument only applies to aircraft flying at low subsonic speeds.

APPENDIX III

AMBIENT NOISE LEVELS

The maximum allowable basic noise level for residential outdoor spaces [8, 9] is 45 dB(A). The appropriate corrections in Table A1 should be applied to this basic noise level to obtain the relevant noise limit referred to as ambient noise level in this report.

TABLE A1

Influencing factor	Correction [dB(A)]
<i>Time period</i>	
(a) Day-time	0
(b) Evening, week-ends	-5
(c) Night-time	-10
<i>Type of district</i>	
(a) Rural residential, hospital zones, recreational	0
(b) Suburban residential, districts with little road traffic	+5
(c) Urban residential	+10
(d) Urban residential with some workshops, business premises or main roads	+15
(e) City (business, trade, administration)	+20
(f) Predominantly industrial area (heavy industry)	+25

APPENDIX IV

GRAPHICAL DETERMINATION OF EMPIRICAL CONSTANTS IN EQUATION (12)

If the following empirical equation is assumed for positive values of t :

$$10^{L_t/10} = 10^{L/10} e^{-a(ut/s')^b}, \quad (\text{A5})$$

then it follows that

$$L_t/10 = L/10 - a(ut/s')^b \log e,$$

$$\therefore a(ut/s')^b = (L - L_t)/10 \log e,$$

$$\therefore \log a + b \log (ut/s') = \log [(L - L_t)/10 \log e]. \quad (\text{A6})$$

Equation (A6) is a linear equation in $\log(ut/s')$ and the slope b and the intercept $\log a$ may be determined by plotting $\log[(L - L_t)/10 \log e]$ against $\log(ut/s')$. These parameters are determined from actual noise-time records by reading off the duration t at arbitrarily chosen values of $(L - L_t)$ and subsequently calculating the values of the parameters.

APPENDIX V

COMPARISON OF THE RESULTS OF DIFFERENT NOISE DURATION EQUATIONS

For the purpose of this comparison, the following have been assumed:

H = aircraft altitude = 300 m,

u = aircraft speed = 120 m s⁻¹,

L_a = ambient noise level = 45 dB(A),

d = ground distance, in m, from flight path,

T_c = noise duration, in s, from computer model (see section 3.2),

T = noise duration, in s, from equation (17),

T_{10} = 10 dB-down duration, in s, from computer model (see section 3.2).

Comparative results are as follows.

TABLE A2

d	T_c	T	T_{10}
0	4.7	4.6	8.9
1500	22.8	23.2	31.2
3000	41.6	43.0	64.0
4500	48.5	58.7	97.0
6000	53.0	62.8	no solution

The discrepancies between T_c and T_{10} for higher values of L_a are much larger than those indicated in the table.